

Rohan Kale

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EDUCATION

The Pennsylvania State University

Aug 2024 – May 2028

B.S. in Computer Science

GPA: 3.55/4.0

Awards/Honors: Dean's List (Top 5% of class)

Relevant Coursework: Computer Architecture, Operating Systems, System Programming, Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Probability & Statistics

EXPERIENCE

CrowdStrike | Software Engineer Intern

May 2026 – Aug 2026

- Designed and implemented a Kafka-based event signaling system in Go, enabling real-time audit event emission across 2,400+ existing customer workflows
- Extended event emission to all firewall mutation paths across 13 REST endpoints and 4 domain entities, generating approximately 18,000 protobuf events/day across 5 cloud regions
- Diagnosed and resolved a data race causing nondeterministic test failures and inconsistent event ordering, enforcing deterministic emission across 15 event sites
- Eliminated nil-dereference panics in customer-facing policy CRUD paths by enforcing non-nil object invariants, removing 27 redundant defensive checks across the codebase

Jetson.app | Software Engineer Intern

June 2024 – August 2024

- Built a React + Node.js quiz system backed by REST APIs that sampled non-repeating questions from a database-backed question bank and persisted user responses and scores for a financial literacy platform serving 10,000+ users
- Accelerated Docker builds by restructuring image layers and caching dependencies across CI runs, reducing pipeline build time from 12 to 7 minutes
- Streamlined randomized question retrieval by indexing frequently queried question-bank fields and consolidating database lookups from 4 queries to 1 per quiz, reducing redundant database access

Horizon Academics | Cybersecurity Research Assistant

June 2023 – August 2023

- Performed packet-level network analysis using Wireshark to examine TCP/IP behavior during simulated denial-of-service attacks
- Generated controlled SYN-flood traffic using hping3 to evaluate the resilience of distributed server architectures under adversarial network conditions
- Investigated and documented DDoS mitigation techniques, including DNS amplification defenses and protocol-level countermeasures, as part of a formal research study

PROJECTS

PrivateAI Gateway | Go, Python, Docker

August 2026

- Engineered an LLM gateway in Go that intercepts prompts and enforces configurable privacy policies, detecting and redacting personally identifiable information (PII) and credentials before requests reach external model providers
- Designed a configurable privacy policy engine that classifies sensitive data and applies allow, redact, tokenize, or block actions, with reversible tokenization for restoring authorized values in model responses
- Implemented asynchronous request processing and streaming model responses, allowing the gateway to handle concurrent inference requests without blocking on downstream model generation

Kale Space Program | Rust, wgpu, winit

June 2026

- Developed a Rust-based 3D planetary simulation engine using wgpu and GPU compute shaders, simulating 20,000+ celestial bodies in real time at 60 FPS through GPU-accelerated physics and instanced rendering
- Accelerated gravitational force computation using WGSL compute shaders and Barnes-Hut spatial approximation, reducing physics-step latency from 62 ms to 14 ms at 5,000 bodies
- Designed a GPU-resident physics pipeline using storage buffers and bind groups to update thousands of body states in parallel while minimizing per-frame CPU-GPU synchronization and data transfers

SKILLS

Languages: Go, Rust, Python, JavaScript, SQL, WGSL, Java, C++

Infrastructure & Data: Kafka, PostgreSQL, Docker, Linux, SQLite, Kubernetes, AWS, GCP, Redis, Protocol Buffers, REST APIs, CI/CD

Frameworks & Tools: FastAPI, Flask, Node.js, React, wgpu, winit, Git, GitHub, BitBucket, Github Actions, Jenkins, Wireshark, hping3, Claude, Codex